Lead Scoring Summary Report

Introduction

This project focused in developing Lead Scoring Model using Logistic Regression to identify potential customers who are most likely to convert. The main of this assignment was to help the sales and marketing team focus more on high-quality leads and improve overall business efficiency. Lead scoring helps business reduce unnecessary effort and prioritize customers who show a higher probability of conversion.

Data Preprocessing

The dataset included customer interaction details, website activity , and lead-related information. We completed several preprocessing steps before building the model. Missing values were identified and handled properly, and unnecessary columns were removed from the dataset. Categorical variables were converted into numerical format using dummy variable creation. Feature scaling was also applied using StandadrdScaler, and the dataset was divided into training and testing data for model evaluation.

Model Building and Evaluation

The regression model was built using Python libraries such as Pandas, NumPy, and Scikit-Learn in Jupyter Notebook. Model was trained using the training dataset, and predictions were generated on the testing dataset. The model performance was evaluated using accuracy score, classification report and confusion matrix. The model achieved an accuracy of approximately 45% on the test data.

Key Findings

In the analysis we saw that variables such as Total Time spent on Website , Total Visits, And Lead Source had a strong impact on Lead conversion probability. Customers who spent more time on the

website and interacted more frequently had a higher chance of conversion. These findings can help the sales team focus on more valuable leads and improve overall marketing performance.

Learnings and Conclusion

This project gave me a solid understanding of the entire machine learning process. Now I feel confident in handling data preprocessing feature engineering, and model evaluation. It has also improved practical knowledge of Logistic Regression and data analysis techniques. In summary , the project successfully demonstrated how lead scoring can support business decision-making and help organizations improve their sales strategies.